MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT: CKU-5/A/A REWORK ASSEMBLY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirement for the refurbishment of the CKU-5/A/A and CKU-5/A rocket catapults which includes renovation of salvageable hardware, regraining, and replacement of non-repairable components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-82733	Cartridge, Impulse, CCU-22/A/A Assembly
MIL-C-82734	Catapult, Aircraft Ejection Seat, CKU-5/A/A Assembly

STANDARDS

MILITARY

MIL-STD-413	Visual Inspection Guide for Elastomeric O-Rings
MIL-STD-453	Inspection, Radiographic
MS17983	Compass-Magnetic, Pilots Standby

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Sea Systems Command (Code Ident 53711)

DL 5184322	Catapult, Aircraft Ejection Seat, CKU-5/A/A Assembly
PL 5184328	Catapult, Aircraft Ejection Seat, CKU-5/A/A Metal Parts Assembly
5184319	Tube, Booster
5184322	Catapult, Aircraft Ejection Seat, CKU-5/A/A Assembly
5184323	Cartridge, Impulse, CCU-22/A/A Assembly
5184328	Catapult, Aircraft Ejection Seat CKU-5/A/A Metal Parts Assembly

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD shall be issued as listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1-78	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subject to first article inspection (see 4.3).

3.2 Conformance to documents. Materials, parts and assemblies, reclaimed and new, shall comply with the requirements specified on DL 5184322, applicable drawings, and referenced documents.

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3.3 Catapult receipt, inspection, and disposition. The following procedures shall be followed:

3.3.1 Government furnished material. Government furnished material (GFM) shall be inspected during the disassembly process to ascertain reclaimable components are not damaged.

3.3.2 Reuseable components. The components listed in TABLE I are expected to have a 90 percent recovery rate and be retained for modification and/or reuse.

TABLE I. Reusable components.

Component	Number required	Drawing/part number
Tang 1/	1	11726657
Plug	1	11744253
Breech 1/	1	11743979
Nozzle	1	11743978
Tube, Booster	1	11726656
Tube, Launcher, Assy	1	11726800
Tube, Motor	1	17726650
Plug, Breech	4	5184315
Tube	1	11744249

1/ Repair threads as necessary

3.3.3 New components. The reworked catapult assembly shall contain new components as listed in TABLE II.

TABLE II. Required new components.

Component	Number required	Reference/Drawing
Screw	1	MS35266-79
O-ring	1	988AS150-2 (AS568-010)MS29513-010
O-ring	4	988AS150-2 (AS568-020)MS29513-020
O-ring	2	988AS150-2 (AS568-024)MS29513-024
O-ring	As Required	988AS150-2 (AS568-022)MS29513-022
O-ring	1	988AS150-2 (AS568-014)MS29513-014
O-ring	2	988AS150-7 (AS568-228)ZZR-765B-228
O-ring	1	988AS150-2 (AS568-026)MS-29513-026
O-ring	1	988AS150-2 (AS568-120)MS-29513-120
O-ring	4	MS28788-4
Set Screw, Hexagon Socket	1	MS51963-37
Set Screw, Hexagon Socket	1	MS51963-34
Cup	1	11726799
Seal	1	11726798
Pin	2	11744252
Cap, Protective	1	11740596
Lockwire	2	11735367
Pin, Spring	2	MS171522
Spring, Tang	1	11726660
Retainer	1	11744248
Head	1	11744255

3.3.3.1 Nozzle. Nozzles conforming to Dwg. 11743978 revision (E) and subsequent revisions may be used in the rework assembly. All others must be replaced.

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3.3.3.2 Cartridge Assembly. The cartridge assembly shall be manufactured in accordance with Dwg. 5184323 and shall comply with the requirements of MIL-C-82733. Only new CCU-22/A/A impulse cartridges manufactured within 12 months prior to the contract scheduled delivery date for the catapult assembly lot shall be installed in the CKU-5/A/A rework assembly (Dwg. 5184322).

3.3.4 Propellant. The propellant used shall be manufactured within 150 days of the contract scheduled delivery date for the reworked catapult assembly lot.

3.3.5 Modified components. Conversion of CKU-5/A catapults and CKU-5/A/A catapults to the latest CKU-5/A/A catapult configuration shall be accomplished by modifying the booster tube component as follows:

3.3.5.1 Booster tube. Modify the booster tube which has a single o-ring groove configuration in accordance with Dwg. 11726656 Revision B, to the dual o-ring groove configuration conforming to Dwg. 5184319. New production booster tubes which have dual o-ring grooves and conform to Dwg. 11726656 may also be used in the rework assembly.

3.4 Hydrostatic pressure. The components listed in TABLE III shall withstand the designated internal pressure for 15 seconds (s) minimum, without application of any external load, and shall thereafter show no evidence of leakage or mechanical failure.

TABLE III. Internal pressure requirements.

Component	Hydrostatic pressure, psi, minimum
Booster tube (11726656)	12,000
Launcher tube (11726800)	12,000
Motor tube (11726650)	5,000

3.5 Protective finish. Protective finish shall be in accordance with the applicable specification referenced on the drawings and shall apply to all surfaces regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied. The tang P/N 11726657 shall have the specified passivation treatment renewed (see 6.4).

3.6 Residual magnetism. The assembled catapult shall not deflect the indicator of an MS17983 compass more than 5° in either direction when passed in front of the compass at the distance of 6 inches.

3.7 Torque. The booster tube and breech assembly, when seated with a torque of 400+50.-0 inch-pounds, shall not be loosened or deformed when subjected to a break-away torque of 200 inch-pounds.

3.8 Ballistics. The catapult, loaded with acceptable explosive components, shall function without mechanical failure and shall meet the ballistic requirements of MIL-C-82734.

3.9 Radiographic requirements. Each loaded rocket catapult shall be radiographically inspected for the presence, condition, and correct placement of internal components.

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3.10 Workmanship. The workmanship shall be that required by the applicable drawings, referenced specifications and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the unit or components, or be a safety hazard in handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65 to 95°F
b. Altitude: Normal ground
c. Vibration: Not applicable
d. Humidity: Room ambient to 95 percent relative, maximum

4.3 First article inspection. Unless otherwise specified in the contract (see 6.2), a first article sample shall be subjected to and shall meet all the first article quality assurance provisions and requirements specified in MIL-C-82734.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the examinations of 4.5 and the tests of 4.6 performed on samples selected in accordance with 4.4.2 and 4.4.3, respectively.

4.4.1 Lot size. For the purpose of identification and shipment, a shipping lot shall consist of 360 rocket catapults, maximum, exclusive of test samples, provided the homogeneity of the manufacturing process has been retained throughout the inspection lot unless otherwise specified (see 6.2).

4.4.1.1 Identification marking for breech. Unless otherwise specified (see 6.2), each breech shall have the previous serial number stamped out and a new serial number engraved on the bottom side of the breech.

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4.4.2 Sampling for examinations. One hundred percent examination shall be performed for all critical defects. Sampling for major and minor defects shall be in accordance with TABLE IV.

TABLE IV. Sampling plan and individual defect criteria.

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1-50	100% examination	0	1	1	2
51-90	50	0	1	2	3
91-150	60	1	2	3	4
151-280	75	1	2	5	6
281-360	95	2	3	7	8

4.4.3 Sampling for tests. Sampling for tests shall be in accordance with the following:

4.4.3.1 Hydrostatic pressure. All units of each component listed in TABLE III shall be subjected to the internal hydrostatic pressure test of 4.6.1.

4.4.3.2 Protective finish. The number of specimens and acceptable criteria for projective finish tests shall be in accordance with the applicable documents listed on the drawings (see 4.6.2).

4.4.3.3 Residual magnetism. All catapult assemblies shall be subjected to the residual magnetism test of 4.6.3.

4.4.3.4 Breakaway torque. Each booster tube and breech assembly shall be subjected to the breakaway torque test of 4.6.4.

4.4.3.5 Ballistics. Sampling for, and lot rejection resulting from, the ballistics test of 4.6.5 shall be as specified in MIL-C-82734.

4.4.3.6 Radiographic inspection. All catapult assemblies shall be subjected to the radiographic inspection of 4.6.6. All failures shall be rejected.

4.5 Quality conformance examination. Using the sampling plan and associated criteria in TABLES IV and V, the quality conformance examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained in TABLE VI. For an individual defect (single characteristic), the criteria in TABLE IV shall be used. For total defects within a major or minor class, the criteria of TABLE V shall be used. All nonconforming material shall be removed from the lot. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Any catapult assembly with a critical defect shall be rejected.

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TABLE V. Total defect criteria.

Number of defect classifications in class 1/	Sample Size 50	60	75	95
	Major defects allowed			
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
	Minor defects allowed			
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

1/ The maximum number of major defects allowed is 14 and the maximum number of minor defects allowed in 20.

4.5.1 Reclaimed parts. Examination of reuseable parts, which required no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All conforming parts shall be reprocessed or rejected.

4.5.2 Reprocessed parts. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained in TABLE VI, using the sampling plan and associated criteria in TABLES IV and V. For an individual defect (single characteristic), the criteria in TABLE IV shall be used. For total defects within a major or minor class, the criteria in TABLE V total defects within a major or minor class, the criteria in TABLE V shall be used. All nonconforming material and parts shall be rejected.

TABLE VI. Classification of defects.

Classification	Defects	Inspection Method
Tube, Motor, P/N 11726650		
Major		
M101	Metal Defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs or sharp corners	Visual

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TABLE VI. Classification of defects (Cont.).

Classification	Defects	Inspection Method
Tube, Booster, P/N 11726656 (rework only)		
Major		
M101	Metal Defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs or sharp corners	Visual
M104	Foreign material in threads	Visual
Tube, Booster, P.N 5184319		
Major		
M101	Length from end of tube to second o-ring groove outside specification	Gage
M102	Width of second o-ring groove inadequate	Gage
M103	Diameter of second o-ring groove inadequate	Gage
M104	Concentricity of second o-ring groove inadequate	Open set-up
M105	Material defective	Visual
M106	Protective finish inadequate or defective	Visual
M107	Burrs or sharp corners	Visual
M108	Foreign matter	Visual
M109	Foreign material in threads	Visual
Tang, P/N 11726657		
Major		
M101	Metal defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs or sharp corners	Visual
M104	Foreign matter	Visual
M105	Foreign material in threads	Visual
Tube, Launcher Assy, P/N 11726800		
Major		
M101	Metal defective	Visual
M102	Wrap defective	Visual
M103	Tube having corrosion or surface damage	Visual

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TABLE VI. Classification of defects (Cont.).

Classification	Defects	Inspection Method
Breech, P/N 11743979		
Critical - Previous lot and serial number stamped out.		
Major		
M101	Metal defective	Visual
M102	Protective finish inadequate	Visual
M103	Burrs or sharp corners	Visual
M104	Foreign material	Visual
M105	Threads not conforming to .4375-20UNJF-3B pitch diameter requirement (maximum)	Gage
M106	Threads not conforming to .4375-20UNJF-3B minor diameter requirement (maximum)	Gage
Minor		
201.	Threads not conforming to 1.125-12UNF-2B pitch diameter requirement	Gage
202.	Threads not conforming to 1.125-12UNF-2B minor diameter requirement	Gage
Nozzle, P/N 11743978 1/		
Major		
M101	Metal defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs or sharp corners	Visual
M104	Foreign matter	Visual
Plug, P/N 11744253		
Major		
M101	Metal defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs or sharp corners	Visual
M104	Foreign matter	Visual
Tube, P/N 11744249		
Major		
M101	Metal defective	Visual
M102	Protective finish inadequate or defective	Visual
M103	Burrs and sharp corners	Visual

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TABLE VI. Classification of defects (Cont.).

Classification	Defects	Inspection Method
M104	Foreign matter	Visual
M105	.031 ± .005 diameter holes (8 places) blocked	Visual

1/ Nozzles from lots 03-IHM-1077 thru IH-81F001-005, and commercial lots of 04-UPC-1178 thru UPC-81B002-002 are to be replaced with new nozzles conforming to drawing 11743978. The following inspection procedure pertains only to pieces from the subsequent lots.

4.6 Test methods and procedures.

4.6.1 Hydrostatic pressure. Prior to the application of the protective finish, each component listed in TABLE III shall be individually mounted in a test device, using the fixture for sealing all openings and subjected to the applicable internal hydrostatic pressure specified in TABLE III to determine compliance with 3.4. The test device used shall internally seal the components within 2.5 inches of each end without applying an axial load other than that applied by the liquid pressure. Any evidence of leakage or mechanical failure shall be cause for rejection of that component.

4.6.2 Protective finish. Test procedures and acceptance criteria for protective finish shall be in accordance with the applicable documents referenced on the drawings.

4.6.3 Residual magnetism. Each catapult assembly shall be slowly passed 6 inches in front of an MS17983 compass along the entire length of the catapult in accordance with FIGURE 1. If the compass indicator deflects more than 5° in either direction, the catapult assembly shall be rejected. The test shall be performed in an area that is shielded from local magnetic effects. When the reworking and loading of an assembly are performed by the same facility, the residual magnetism requirement shall be performed on the loaded assembly only.

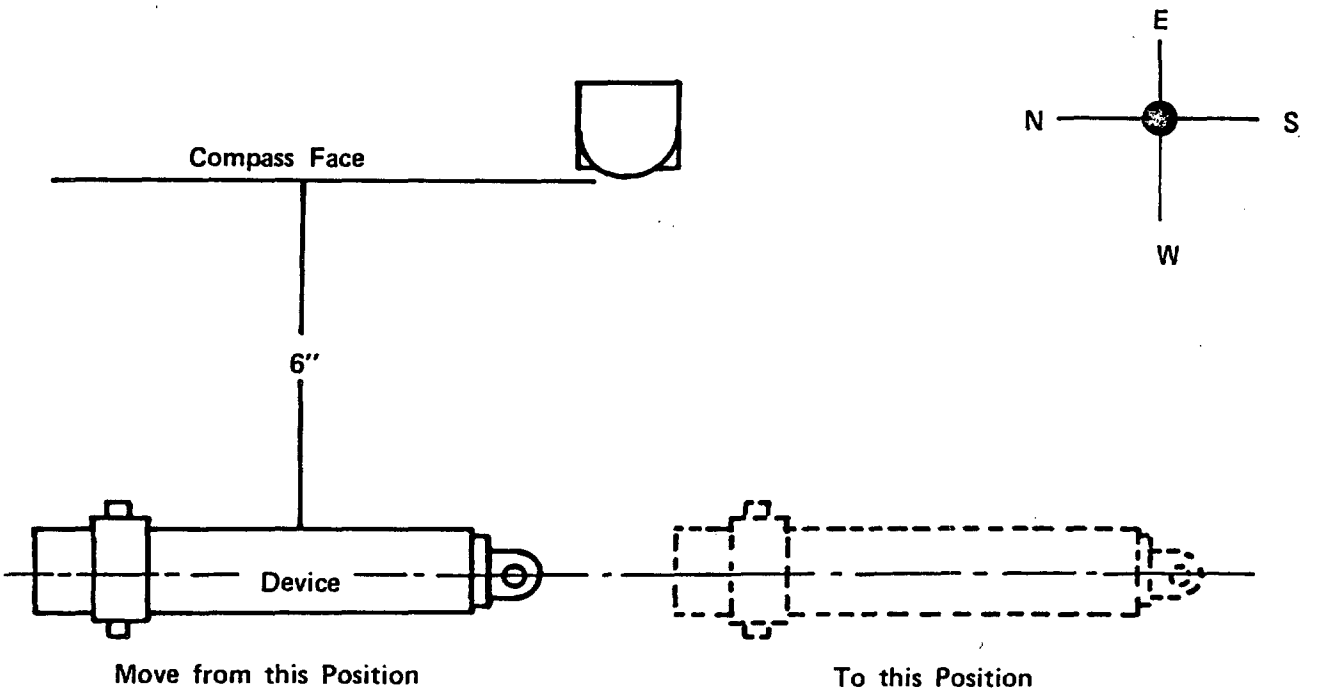
4.6.4 Breakaway torque. A torque wrench set at the torque requirements specified in 3.7 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque. Any evidence of loosening or deformation of any component shall result in rejection of the catapult assembly.

4.6.5 Ballistics. The test samples shall be tested to the ballistic requirement of MIL-C-82734.

4.6.6 Radiographic inspection. Rocket catapults shall be radiographically inspected in accordance with MIL-STD-453, Radiographic Quality Level I, at 0° and 90° positions. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket catapult components 0.03 inch in length or width.

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Pilots Standby Compass MS17983



Plan View

Note: Device and Compass Shall Lie in a Common Horizontal Plane

FIGURE 1. Layout for determination of residual magnetism in device.

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with Dwg. 5184320.

6. NOTES

6.1 Intended use. The aircraft ejection seat catapult metal parts assembly composed of new, reclaimed, and modified components as applicable and as described herein and when loaded with propellant, in accordance with MIL-P-82663 and drawing 11744246 and a CCU-22/A/A impulse cartridge, MIL-C-82733, will produce the catapult CKU-5/A/A assembly. The catapult assembly is to meet the requirements of MIL-C-82734 and is intended to be used in conjunction with other propellant actuated devices in aircrew emergency escape systems used in jet aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity required.
c. Inspection conditions when other than as specified (see 4.2.1).
d. When a first article sample is not required (see 4.3).
e. Exceptions to lot homogeneity (see 4.4.1).
f. Breech identification marking if other than as specified (see 4.4.1.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of the number of units specified for first article inspection in MIL-C-82734. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the documents first articel.

6.4 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part or evidence of leakage, the occurrence of which is other than a design function.

6.5 Remanufactured exterior components. Remanufactured exterior components are as follows:

a. Motor Tube, Dwg. 11726650
b. Breech, Dwg. 11743979
c. Nozzle, Dwg. 11743978
d. Tang, Dwg. 11726657

Custodian:
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Preparing Activity:
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Project Number:
1377-N747

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-82735(OS)	Catapult, Aircraft Ejection Seat: CKU-5/A/A Rework Assembly

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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